DBT PORTFOLIO PROJECT

LAYER: INTERMEDIATE

DBT Intermediate Model

Sharpe Ratio | Risk-Adjusted Return | Finance | Statistics | Snowflake

Project Summary

Computes the Sharpe Ratio -- the industry-standard measure of risk-adjusted return -- for each entity/payer/state/specialty segment. Subtracts a monthly risk-free rate of 0.4167% (5% annualized / 12) from each period's return, divides by standard deviation of excess returns, then annualizes by multiplying by sqrt(12). Feeds directly into the mart layer and Tableau risk-adjusted performance dashboards.

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1 PROJECT OVERVIEW

Model Name:	int_sharpe_ratio
Layer:	Intermediate
Materialization:	table
Risk-Free Rate:	0.004167 per period (5% annual / 12 months)
Key Outputs:	sharpe_periodic, sharpe_annualised per entity/payer/state/specialty
Grain:	One row per entity_id / payer / state / specialty
Audience:	CFO, Finance leadership, Investment committee, Tableau dashboards

2 BUSINESS PROBLEM

Two payers may have similar average returns, but one achieves that return with half the volatility. The Sharpe Ratio surfaces this -- higher Sharpe = better return per unit of risk. Finance leadership needed this metric segmented by payer, state, and specialty to make capital allocation and contract negotiation decisions backed by risk-adjusted data, not raw return figures.

- sharpe_periodic: Period-level Sharpe -- comparable across entities within the same time horizon.
- sharpe_annualised: Multiplied by sqrt(12) to convert monthly Sharpe to annual convention.
- NULLIF(sigma_p, 0): Guards against division by zero for entities with constant returns.
- Grouped by payer, state, specialty: Enables segment-level ranking and benchmarking.
- Risk-free rate of 0.004167: Represents ~5% annualized Treasury rate divided by 12 periods.

3 TECHNICAL EXPLANATION

- excess CTE: Subtracts the monthly risk-free rate (0.004167) from each period's return_value to compute excess_return per row.
- agg CTE: Groups by entity/payer/state/specialty -- computes AVG(excess_return) as mean_excess_return and STDDEV_SAMP(excess_return) as sigma_p. COUNT(*) tracks n_periods.
- sharpe_periodic = mean_excess_return / sigma_p -- the core Sharpe formula.
- sharpe_annualised = sharpe_periodic * sqrt(12) -- annualization assuming 12 monthly periods.
- Materialized as TABLE: Referenced by mart layer alongside VaR -- table avoids re-aggregation.

Key Techniques: Excess return computation, AVG(), STDDEV_SAMP(), NULLIF division guard, sqrt(12) annualization

4 FULL MODEL CODE

```
{{ config(materialized='table') }}

with excess as (
    select
        entity_id,
        payer,
        state,
        specialty,
        return_value - 0.004167 as excess_return
    from {{ source('healthcare', 'claims') }}
),

agg as (
    select
        entity_id,
        payer,
        state,
        specialty,
        avg(excess_return)        as mean_excess_return,
        stddev_samp(excess_return) as sigma_p,
        count(*)                  as n_periods
    from excess
    group by entity_id, payer, state, specialty
)

select
    *,
    mean_excess_return / nullif(sigma_p, 0)                    as sharpe_periodic,
    (mean_excess_return / nullif(sigma_p, 0)) * sqrt(12)       as sharpe_annualised
from agg
```

5 EXAMPLE OUTPUT

entity_id	payer	state	mean_excess	sigma_p	n	sharpe_periodic	sharpe_annual
E-001	BlueCross	AR	0.0278	0.0180	120	1.544	5.347
E-002	Aetna	TX	0.0148	0.0240	118	0.617	2.136
E-003	UnitedHlth	MO	0.0368	0.0120	122	3.067	10.621

** Sharpe > 1.0 periodic is generally considered strong. Risk-free rate = 0.4167% per period (5% annualized).*